

Decision Function-Based Adaptive Control of Uncertain Systems Subject to Irregular Output Constraints

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Abstract—While barrier Lyapunov function-based approach has been successfully used in adaptive control of uncertain systems with output/state constraints, it is nontrivial to use this method to deal with systems with irregular constraints (where the constraints may appear, disappear, or alternate during system operation). It is in fact an open interesting problem to develop a control method capable of coping with the irregular constraints uniformly. In this note, a unified adaptive control strategy is developed for uncertain systems subject to irregular output constraints. First, we introduce the concept of link function, with which we transform the nondifferentiable constraint into a smooth one. Second, by using several function transformations, and the decision function in particular that bridges the irregular output constraints and the barrier function, we recast the problem of maintaining irregular constraints into the stabilization of the barrier function, which allows different constraint forms to be addressed under a fixed control structure wherein one only needs to select the appropriate decision function and design parameters, eliminating the need for control redesign and stability reanalysis. The effectiveness of the proposed scheme is validated by numerical simulations.

Index Terms—Adaptive control, backstepping, decision function, output constraints, uncertain systems.

I. INTRODUCTION

Driven by practical needs and safety considerations, the controlled uncertain plant may inevitably suffer from output/state constraints [1], [2]. Therefore, it is normally required for the corresponding controls to be able to maintain certain constraints, rather than just closed-loop stability. However, under different working environments, uncertain systems may be subject to various forms of constraints, which can be mainly divided into the following cases.

- 1) *Type A (Unconstrained form)*: There is no constraint during the system operation, which is denoted by “ C_{None} ” and the constraining boundary can be viewed as infinity.
- 2) *Type B (Persistent constrained form)*: The constraints are always present during system operation, which is denoted by “ $C_{[0,\infty)}$ ”.

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- 3) *Type C (Free-Constrained form)*: There is no constraint initially and later the constraints are imposed at $t = T_1$ and thereafter, which is denoted by “ $F_{[0,T_1)} C_{[T_1,\infty)}$ ”.
- 4) *Type D (Constrained-Free form)*: The constraints are imposed from the beginning of system operation and then are removed at $t = T_1$, which is denoted by “ $C_{[0,T_1)} F_{(T_1,\infty)}$ ”.
- 5) *Type E (Alternating Free-Constrained form)*: The free-constrained form appears alternately during system operation, which is denoted by “ $F_{[0,T_1)} C_{[T_1,T_2)} F_{(T_2,T_3)} \dots$,” where T_k , $k = 0, 1, 2, \dots$, denotes the constrained/unconstrained switching time with $T_0 = 0$.
- 6) *Type F (Alternating Constrained-Free form)*: The constrained-free form appears alternately during system operation, which is denoted by “ $C_{[0,T_1)} F_{(T_1,T_2)} C_{[T_2,T_3)} \dots$ ”.

For the Type A form, it means that there are no output/state constraints during system operation, there are many fruitful results in the literature; see for examples, [3], [4], and [5]. Inspired by the pioneering work [6], barrier Lyapunov function (BLF)-based adaptive controls were proposed for uncertain strict-feedback systems to handle the constraint with Type B form (see [7], [8], [9], [10], [11], [12], [13], and the references therein), in which by employing different kinds of BLF, the problems of constant/time-varying constraints with symmetric/asymmetric forms are solved. However, it is worth noting that the above BLF-based results are based on the assumption that the output/state constraints are imposed from the beginning of system operation and are maintained for the whole system operation, namely, only the Type B constrained form can be handled. Recently, by making some modifications to the barrier function, the constrained algorithms in [14] and [15] can handle Type A and Type B forms uniformly. In practice, however, there are many other forms of constraints (i.e., Type C–Type F). In other words, the commonly encountered constraints are not Type B and may be imposed only for a specified time (rather than all the time) or may be removed at specified times during system operation. A typical example of such constraints is that a mobile robot runs in a free-moving area for a while and then enters a constrained area, for example to avoid collisions or to hide a target [9], [16]. Thus, how to handle the general yet irregular constraints denotes an interesting work.

Technically, for constrained systems with Type C–Type F forms, since the constraint boundaries are sometimes infinity and sometimes bounded, the bounds themselves are discontinuous and nondifferentiable, which means that the regular BLF-based adaptive control algorithm is no longer applicable for uncertain strict-feedback systems, as the traditional backstepping method requires that the constraint functions must be differentiable. Motivated by the work in [17], the work [18] was the first time to propose an adaptive control algorithm for strict-feedback uncertain systems that guarantees Type C constraint. Subsequently, a “free-constrained-free” constraint form has been solved in [19] by improving the shifting function. However, it is worth noting that the corresponding controls in [18] and [19] lack the generalization, i.e., only some special forms can be handled and other scenarios cannot be coped with uniformly. If the designers would like to tackle other constraint forms, control redesign and stability

reanalyze must be implemented, which makes the control algorithm less user-friendly. Therefore, how to design a unified control approach for uncertain nonlinear systems, capable of coping with the Type A–Type F constraints, is still an open problem.

In this note, we propose an adaptive control scheme for uncertain systems subject to irregular output constraints. The main contributions and novelties of this note are summarized as follows.

- 1) It is the first time that a systemic design framework for parametric uncertain systems is developed so that multiple constraints (i.e., Type A–Type F) can be handled in a unified way, which is more powerful and flexible compared with the existing BLF-based methods [8], [9], [10], [11], [12].
- 2) Since the Type C–Type F constraints are discontinuous, the original backstepping technique is no longer available. To solve this problem, we propose a link function and a soft function here so that the discontinuous constraints can be transformed into the smooth forms, making the constraint function differentiable.
- 3) To handle the constraints with Type A–Type F uniformly, a set of function transformations (i.e., decision function, soft function, and barrier function) is developed. It is revealed from Lemma 1 that as long as the boundedness of the barrier function is guaranteed, the six constraint types can be preserved by the proper choice of decision functions and key parameters, so that not only the control redesign is not required but also the applicability and generality of the proposed control are strengthened.

II. CONSTRAINTS AND TRANSFORMATIONS

Consider an uncertain nonlinear system¹

$$\begin{cases} \dot{x} = f(x, u, \theta), & x(0) = x_0 \\ y = \mathcal{X} \end{cases} \quad (1)$$

where $x \in \mathbb{R}^m$ is the state vector with x_0 being the initial state, $u \in \mathbb{R}^p$ is the control input, $\theta \in \mathbb{R}^r$ is an unknown system parameter, $f : \mathbb{R}^m \times \mathbb{R}^p \times \mathbb{R}^r \rightarrow \mathbb{R}^m$ is a smooth function, and $y = \mathcal{X} \in \mathbb{R}^q$ is the system output that is continuously differentiable. Moreover, the system output is subject to various constraint forms with Type A–Type F. For simplicity, without loss of generality, we consider the case $q = 1$ in the discussion of the constraints in Section I.

A. Hard Constraint Forms

The six constraint forms (Type A–Type F) discussed in Section I can be mathematically written as follows:

- 1) Type A (C_{None}): $-\infty < \mathcal{X}(t) < \infty \quad \forall t \geq 0$;
- 2) Type B ($C_{[0, \infty)}$): $\mathcal{F}_1^1(t) < \mathcal{X}(t) < \mathcal{F}_2^1(t) \quad \forall t \geq 0$;
- 3) Type C ($F_{[0, T_1]} C_{[T_1, \infty)}$):

$$\begin{cases} -\infty < \mathcal{X}(t) < \infty & \forall t \in [0, T_1) \\ \mathcal{F}_1^1(t) < \mathcal{X}(t) < \mathcal{F}_2^1(t) & \forall t \in [T_1, \infty); \end{cases} \quad (2)$$

- 4) Type D ($C_{[0, T_1]} F_{(T_1, \infty)}$):

$$\begin{cases} \mathcal{F}_1^0(t) < \mathcal{X}(t) < \mathcal{F}_2^0(t) & \forall t \in [0, T_1] \\ -\infty < \mathcal{X}(t) < \infty & \forall t \in (T_1, \infty); \end{cases} \quad (3)$$

- 5) Type E (i.e., $F_{[0, T_1]} C_{[T_1, T_2]} F_{(T_2, \infty)}$):

$$\begin{cases} -\infty < \mathcal{X}(t) < \infty & \forall t \in [0, T_1) \\ \mathcal{F}_1^1(t) < \mathcal{X}(t) < \mathcal{F}_2^1(t) & \forall t \in [T_1, T_2] \\ -\infty < \mathcal{X}(t) < \infty & \forall t \in (T_2, \infty); \end{cases} \quad (4)$$

- 6) Type F (i.e., $C_{[0, T_1]} F_{(T_1, T_2)} C_{[T_2, \infty)}$):

$$\begin{cases} \mathcal{F}_1^0(t) < \mathcal{X}(t) < \mathcal{F}_2^0(t) & \forall t \in [0, T_1] \\ -\infty < \mathcal{X}(t) < \infty & \forall t \in (T_1, T_2) \\ \mathcal{F}_1^2(t) < \mathcal{X}(t) < \mathcal{F}_2^2(t) & \forall t \in [T_2, \infty) \end{cases} \quad (5)$$

where $\mathcal{X} \in \mathbb{R}$ denotes the constrained variable, T_i , $i = 0, 1, 2$, is the constrained/unconstrained switching time with $T_0 = 0$, and $\mathcal{F}_1^i \in \mathbb{R}$ and $\mathcal{F}_2^i \in \mathbb{R}$ are bounded constraint functions or constants.

Remark 1: In practical applications, there are many alternating combinations of “free” and “constraint,” i.e., $C_{[0, T_1]} F_{[T_1, T_2]} C_{(T_2, T_3)} F_{[T_3, \infty)}$, which can be expressed as

$$\begin{cases} \mathcal{F}_1^0(t) < \mathcal{X}(t) < \mathcal{F}_2^0(t) & \forall t \in [0, T_1] \\ -\infty < \mathcal{X}(t) < \infty & \forall t \in (T_1, T_2) \\ \mathcal{F}_1^2(t) < \mathcal{X}(t) < \mathcal{F}_2^2(t) & \forall t \in [T_2, T_3] \\ -\infty < \mathcal{X}(t) < \infty & \forall t \in (T_3, \infty) \end{cases} \quad (6)$$

where $\mathcal{F}_1^i$ and $\mathcal{F}_2^i$, $i = 0, 2$, are the constraint boundaries. Here, it should be emphasized that, to facilitate the labeling and to avoid confusion, i depends on the subscript of start time T_i of each constraint interval. Furthermore, (4) and (5) are the simplest forms of Type E and Type F, thus in this note, we focus on handling these simple forms, other alternating forms can be handled similarly.

To summarize, six constraint forms can be uniformly expressed as: $\mathcal{F}_1(t) < \mathcal{X}(t) < \mathcal{F}_2(t) \forall t \in [0, \infty)$, where $\mathcal{F}_1 \in \mathbb{R}$ and $\mathcal{F}_2 \in \mathbb{R}$, denoting infinity or bounded values, are constraint functions for $\mathcal{X}$.

B. Soft Constraint Forms

It is seen from Type C–Type F constraints shown in (2)–(5) that, due to the hard switch between “constraint” and “free” (i.e., T_1 and T_2), the constraint functions $\mathcal{F}_1(t)$ and $\mathcal{F}_2(t)$ are discontinuous and nondifferentiable over the interval $[0, \infty)$. Therefore, unlike the existing works dealing with the persistent constraint [8], [9], [10], [11], [12], [13], it is difficult to directly utilize the adaptive technique to guarantee the constraints. Therefore, in the following we construct a “link function” whose purpose is to ensure the continuity of the constraint function and to transform the hard constraint into a soft constraint. We define two time variables T_i^l and T_i^r satisfying $T_{i-1} < T_i^l < T_i < T_i^r < T_{i+1}$ for $i = 1, 2, \dots$, where T_{i-1} and T_{i+1} represent the last and the next switching times. It is worth emphasizing that the selection of T_i^l and T_i^r can be arbitrarily close to the time moment T_i .

Definition 1: By “link function,” it refers to a class of real functions $l_1(t)$ and $l_2(t)$, satisfying the following features.

- 1) $l_1(t)$ and $l_2(t)$, satisfying $l_1(t) < l_2(t)$, are continuously differentiable over the interval (T_{i-1}^l, T_i^l) or (T_i^r, T_{i+1}^r) .
- 2) For “ $F_{[T_{i-1}, T_i]} C_{[T_i, T_{i+1}]}$ ” form, $l_1(t)$ and $l_2(t)$ over the interval (T_i^l, T_i) meet $\lim_{t \rightarrow T_i^l+} l_2(t) \rightarrow +\infty$, $\lim_{t \rightarrow T_i^-} l_2(t) \rightarrow \mathcal{F}_2^i(T_i)$, $\lim_{t \rightarrow T_i^l+} l_1(t) \rightarrow -\infty$, and $\lim_{t \rightarrow T_i^-} l_1(t) \rightarrow \mathcal{F}_1^i(T_i)$.
- 3) For “ $C_{[T_{i-1}, T_i]} F_{(T_i, T_{i+1})}$ ” form, $l_1(t)$ and $l_2(t)$ over the interval (T_i, T_i^r) meet $\lim_{t \rightarrow T_i^+} l_2(t) \rightarrow \mathcal{F}_2^{i-1}(T_i)$, $\lim_{t \rightarrow T_i^r-} l_2(t) \rightarrow +\infty$, $\lim_{t \rightarrow T_i^+} l_1(t) \rightarrow \mathcal{F}_1^{i-1}(T_i)$, and $\lim_{t \rightarrow T_i^r-} l_1(t) \rightarrow -\infty$.

To facilitate the description, we define l_1 as the lower link function and l_2 as the upper link function. By utilizing the concept of link function, we are able to convert the hard form into a soft one. Now, we take the “ $F_{[0, T_1]} C_{[T_1, \infty)}$ ” form as an example to show the detailed process. For such a constraint, as there is no constraint in the interval $[0, T_1)$, then we impose a time moment T_1^l so that the original interval $[0, T_1)$ can be divided into two parts: $[0, T_1^l]$ and (T_1^l, T_1) , the corresponding expression of “ $F_{[0, T_1]} C_{[T_1, \infty)}$ ” in (2) can be rewritten as the soft virtual

¹ $\mathbb{R}$ denotes the set of real numbers, and the initial time t_0 is set as $t_0 = 0$.

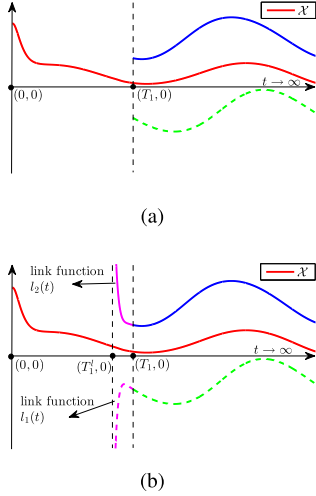


Fig. 1. Schematic illustration of the hard/soft constraint forms. (a) Hard “ $F_{[0, T_1]} C_{[T_1, \infty)}$ ” form. (b) Soft “ $F_{[0, T_1^l]} L_{(T_1^l, T_1^r)} C_{[T_1, \infty)}$ ” form.

form

$$\begin{cases} -\infty < \mathcal{X}(t) < \infty & \forall t \in [0, T_1^l] \\ l_1(t) < \mathcal{X}(t) < l_2(t) & \forall t \in (T_1^l, T_1^r) \\ \mathcal{F}_1^l(t) < \mathcal{X}(t) < \mathcal{F}_2^r(t) & \forall t \in [T_1, \infty) \end{cases} \quad (7)$$

which is denoted by “ $F_{[0, T_1^l]} L_{(T_1^l, T_1^r)} C_{[T_1, \infty)}$ ”. Upon using the definition of link function $l_k(t)$, it is readily shown that the constraint function is continuous everywhere for $\forall t \geq 0$. A brief schematic illustrations of “ $F_{[0, T_1]} C_{[T_1, \infty)}$ ” and “ $F_{[0, T_1^l]} L_{(T_1^l, T_1^r)} C_{[T_1, \infty)}$ ” forms are shown in Fig. 1.

According to the above discussion, it is seen that the original four hard constraints (Type C–Type F) can be converted into the soft virtual constraints and also can be described as a form that similar to (7), i.e.,

- 1) “ $C_{[0, T_1]} F_{(T_1, \infty)}$ ” $\Rightarrow$ “ $C_{[0, T_1]} L_{(T_1, T_1^r)} F_{(T_1^r, \infty)}$ ”;
- 2) “ $F_{[0, T_1]} C_{[T_1, T_2]} F_{(T_2, \infty)}$ ” $\Rightarrow$ “ $F_{[0, T_1^l]} L_{(T_1^l, T_1^r)} C_{[T_1^r, T_2]} L_{(T_2, T_2^r)} F_{(T_2^r, \infty)}$ ”;
- 3) “ $C_{[0, T_1]} F_{(T_1, T_2)} C_{[T_2, \infty)}$ ” $\Rightarrow$ “ $C_{[0, T_1]} L_{(T_1, T_1^r)} F_{(T_1^r, T_2^l)} L_{(T_2^l, T_2^r)} C_{[T_2^r, \infty)}$ ”.

C. Transformation Functions

In the above section, by constructing a link function, we have converted the hard and discontinuous constraint form into the soft and continuous form. However, there still remain some key problems: 1) how to utilize the “link function” to handle different and irregular forms uniformly, and 2) how to ensure the differentiability of link function so that the backstepping technique is suitable for output-constrained strict-feedback systems.

To address these issues, we impose a series of powerful functions (i.e., soft function, decision function, and barrier function).

1) Soft Function: First, we construct a soft function $N(\chi)$ as

$$N(\chi) = \frac{\exp(\chi)}{\exp(\chi) + \exp(-\chi)} \quad (8)$$

with

$$\begin{cases} \chi = \frac{1}{t-T_i^l} - \frac{1}{T_i^r-t} & \forall t \in (T_i^l, T_i^r) \\ \text{or } \chi = \frac{1}{T_i^r-t} - \frac{1}{t-T_i^l} & \forall t \in (T_i, T_i^r) \end{cases} \quad (9)$$

for $i = 1, 2, \dots$. It is interesting from (8) and (9) to note that the soft function $N(\chi)$ exhibits the following properties:

- 1) $N(\chi) \rightarrow 1$ as $\chi \rightarrow +\infty$ and $N(\chi) \rightarrow 0$ as $\chi \rightarrow -\infty$;

- 2) $N(\chi) \in (0, 1) \forall \chi \in (-\infty, \infty)$;

- 3) N and its k th time derivatives are smooth and bounded, $\forall t \in (T_i^l, T_i^r)$ (or $\forall t \in (T_i, T_i^r)$), $k = 1, 2, \dots$;

- 4) $\lim_{\chi \rightarrow \pm\infty} N^{(k)}(t) \rightarrow 0$ for $k = 1, 2, \dots$

2) Pool of Decision Function: Definition 2: A real-time function $\kappa : [0, \infty) \rightarrow [0, 1]$ is called a “decision function” if 1) $\kappa^{(j)}(t)$, $j = 0, \dots, n$, are bounded, known, and piecewise continuous, and 2) it has the following choices.

Case ①. $\kappa(t) \equiv 1 \quad \forall t \in [0, \infty)$.

Case ②. $\kappa(t) \equiv 0 \quad \forall t \in [0, \infty)$.

Case ③. $\kappa(t) = \begin{cases} 1, & t \in [0, T_1^l] \\ N(\chi), & t \in (T_1^l, T_1^r) \\ 0, & t \in [T_1, \infty). \end{cases}$

Case ④. $\kappa(t) = \begin{cases} 0, & t \in [0, T_1] \\ N(\chi), & t \in (T_1, T_1^r) \\ 1, & t \in [T_1^r, \infty). \end{cases}$

Case ⑤. $\kappa(t) = \begin{cases} 1, & t \in [0, T_1^l] \\ N(\chi), & t \in (T_1^l, T_1^r) \\ 0, & t \in [T_1, T_2] \\ N(\chi), & t \in (T_2, T_2^r) \\ 1, & t \in [T_2^r, T_3] \\ \vdots \end{cases}$

Case ⑥. $\kappa(t) = \begin{cases} 0, & t \in [0, T_1] \\ N(\chi), & t \in (T_1, T_1^r) \\ 1, & t \in [T_1^r, T_2^l] \\ N(\chi), & t \in (T_2^l, T_2^r) \\ 0, & t \in [T_2, T_3] \\ \vdots \end{cases}$

Based on the definition of decision function, we construct the following scaling function:

$$\beta = (b_0 - b_f)\kappa + b_f, \quad 0 < b_f < b_0 \quad (10)$$

where b_0 is a positive constant and b_f denotes a positive function/constant. In addition, b_f and its derivatives up to k th, $k = 1, 2, \dots$, are known, bounded, and piecewise continuous.

Remark 2: Upon utilizing the properties of the decision function, it is revealed that β and its derivatives up to k th are also known, bounded, and piecewise continuous for $t \in [0, \infty)$. According to the above selections of decision functions, one has the following.

- 1) For case ①, $\beta(t) \equiv b_0$.

- 2) For case ②, $\beta(t) \equiv b_f$.

- 3) For case ③, $\beta(t) = \begin{cases} b_0, & t \in [0, T_1^l] \\ (b_0 - b_f)N(\chi) + b_f, & t \in (T_1^l, T_1^r) \\ b_f, & t \in [T_1, \infty). \end{cases}$

- 4) For case ④, $\beta(t) = \begin{cases} b_f, & t \in [0, T_1] \\ (b_0 - b_f)N(\chi) + b_f, & t \in (T_1, T_1^r) \\ b_0, & t \in [T_1^r, \infty). \end{cases}$

Remark 3: Upon using the above functions, we give a detailed function that fulfills the properties of the “link function” as

$$\Phi(\beta) = \frac{\lambda\beta}{\sqrt{b_0^2 - \beta^2}} \quad (11)$$

with $\lambda > 0$ and $\lambda^{(k)}$, $k = 1, 2, \dots, n$, being known, bounded, and piecewise continuous. Therefore, the lower and upper link functions

have the following forms:

$$l_1(t) = \Phi(-\beta(t)) = \frac{-\lambda\beta(t)}{\sqrt{b_0^2 - \beta(t)^2}}, \quad l_2(t) = \Phi(\beta(t)) = \frac{\lambda\beta(t)}{\sqrt{b_0^2 - \beta(t)^2}}$$

for $t \in (T_i^l, T_i)$ or $\forall t \in (T_i, T_i^r)$, $i = 1, 2, \dots$. It is easy to verify that the proposed link functions can convert the hard constraints of Type C–Type F into soft and smooth constraints (you may refer to the proof in Lemma 1). It is worth noting that there are many forms similar to (11), for example, $\Phi(\beta) = \tan(\frac{\pi}{2b_0}\beta)$ and $\Phi(\beta) = \frac{\beta}{(b_0+\beta)(b_0-\beta)}$.

3) Barrier Function: In this section, we construct a barrier function to handle different constraint forms uniformly

$$s = \frac{\zeta}{(\underline{\delta} + \zeta)(\bar{\delta} - \zeta)}, \quad \text{with } \zeta = \frac{\eta}{\beta}, \quad \eta = \frac{b_0\mathcal{X}}{\sqrt{\mathcal{X}^2 + \lambda^2}} \quad (12)$$

where $\underline{\delta}$ and $\bar{\delta}$ denote positive constants/functions satisfying

$$0 < \bar{\delta}\beta \leq b_0 \quad \text{and} \quad 0 < \underline{\delta}\beta \leq b_0.$$

In addition, $\underline{\delta}^{(k)}$ and $\bar{\delta}^{(k)}$, $k = 0, 1, \dots, n$, are known, bounded, and piecewise continuous, with n is the order of the considered system.

The following lemma is to explain that why we construct the barrier function.

Lemma 1: For any $\zeta(0)$ satisfying $-\underline{\delta}(0) < \zeta(0) < \bar{\delta}(0)$, if s is bounded for $t \geq 0$, the constraint forms with Type A–Type F can be handled uniformly.

Proof: As $\frac{\partial s}{\partial \zeta} = \frac{\bar{\delta}\bar{\delta} + \zeta^2}{(\bar{\delta} + \zeta)^2(\bar{\delta} - \zeta)^2} > 0$ for $\zeta \in \Omega_\zeta := \{\zeta \in \mathbb{R} \mid -\underline{\delta} < \zeta < \bar{\delta}\}$, it is shown that s is strictly monotonic with respect to (w.r.t.) ζ . Thus, for any initial value $\zeta(0)$ satisfying $\zeta(0) \in \Omega_\zeta$, if s is bounded $\forall t > 0$, it is straightforward that ζ is always within the open set Ω_ζ and there exist some constants $\underline{\zeta}$ and $\bar{\zeta}$ such that

$$-\underline{\delta} < -\underline{\zeta} \leq \zeta \leq \bar{\zeta} < \bar{\delta}. \quad (13)$$

Noting that $\zeta = \frac{\eta}{\beta}$ and $\beta > 0$, it is seen from (13) that $-\underline{\delta}\beta < \eta < \bar{\delta}\beta$. Since $\Phi(\star) = \frac{\lambda\star}{\sqrt{b_0^2 - \star^2}}$ is strictly monotonically increasing for any $\star$ over the set $\Omega_\star := \{\star \in \mathbb{R} \mid \star < b_0\}$, one has

$$\Phi(-\underline{\delta}\beta) < \Phi(\eta) < \Phi(\bar{\delta}\beta). \quad (14)$$

According to the definitions of η shown in (12), it is easy to obtain that $\mathcal{X} = \Phi(\eta)$, then we further have $\Phi(-\underline{\delta}\beta) < \mathcal{X} < \Phi(\bar{\delta}\beta)$, which can be rewritten as

$$\mathcal{F}_1 < \mathcal{X} < \mathcal{F}_2, \quad \text{with } \mathcal{F}_1 = -\frac{\lambda\underline{\delta}\beta}{\sqrt{b_0^2 - (\underline{\delta}\beta)^2}}, \quad \mathcal{F}_2 = \frac{\lambda\bar{\delta}\beta}{\sqrt{b_0^2 - (\bar{\delta}\beta)^2}}. \quad (15)$$

It is observed from (15) that the selection of different decision function κ and parameters $\underline{\delta}$ and $\bar{\delta}$ can result in different constraint forms, as follows.

- 1) For case ①, if $\underline{\delta} = \bar{\delta} = 1$, one has $b_0^2 - (\bar{\delta}\beta)^2 \equiv 0$ and $b_0^2 - (\underline{\delta}\beta)^2 \equiv 0$, then it follows that $\mathcal{F}_2 = \infty$ and $\mathcal{F}_1 = -\infty$, which indicates that there is no constraint on $\mathcal{X}$.
- 2) For case ①, if $0 < \underline{\delta} < 1$ and $0 < \bar{\delta} < 1$, one has $0 < \underline{\delta}\beta = \underline{\delta}b_0 < b_0$ and $0 < \bar{\delta}\beta = \bar{\delta}b_0 < b_0$, it is shown that $\mathcal{F}_2 > 0$ and $\mathcal{F}_1 < 0$ are bounded functions, which implies that there exists a persistent constraint on $\mathcal{X}$.
- 3) For case ②, if $\underline{\delta}$, $\bar{\delta}$, and b_f are chosen so that $\underline{\delta}\beta = \bar{\delta}\beta = b_0$, it is seen that $\mathcal{F}_2 = \infty$ and $\mathcal{F}_1 = -\infty$, which indicates there is no constraint on $\mathcal{X}$.
- 4) For case ②, if $\underline{\delta}$, $\bar{\delta}$, and b_f are selected such that $\underline{\delta}b_f < b_0$ and $\bar{\delta}b_f < b_0$, it is checked that $\mathcal{F}_2 > 0$ and $\mathcal{F}_1 < 0$ are bounded, then there exists a persistent constraint on $\mathcal{X}$.

TABLE I
DIFFERENT CONSTRAINT FORMS UNDER DIFFERENT DECISION FUNCTIONS AND DESIGN PARAMETERS

κ_i	$\bar{\delta}, \underline{\delta}$	Constraint Form
Case ①	$\underline{\delta}, \bar{\delta} = 1$	C_{None}
	$\underline{\delta}, \bar{\delta} < 1$	$C_{[0, \infty)}$
Case ②	$\underline{\delta}b_f = b_0$ $\bar{\delta}b_f = b_0$	C_{None}
	$\underline{\delta}b_f < b_0$ $\bar{\delta}b_f < b_0$	$C_{[0, \infty)}$
Case ③	$\underline{\delta} = 1$ $\bar{\delta} = 1$	$F_{[0, T_1^l]} L_{(T_1^l, T_1)} C_{[T_1, \infty)}$
Case ④		$C_{[0, T_1]} L_{(T_1, T_1^r)} F_{(T_1^r, \infty)}$
Case ⑤		$F_{[0, T_1^l]} L_{(T_1^l, T_1)} C_{[T_1, T_2]} L_{(T_2, T_2^r)} F_{(T_2^r, \infty)}$
Case ⑥		$C_{[0, T_1]} L_{(T_1, T_1^r)} F_{[T_1^r, T_2^l]} L_{(T_2^l, T_2)} C_{[T_2, \infty)}$

- 5) For case ③, if $\underline{\delta} = \bar{\delta} = 1$, it is seen from (15) that, for $t \in [0, T_1^l]$, $\beta = b_0$, one has $\mathcal{F}_2 = -\mathcal{F}_1 = \infty$; for $t \in (T_1^l, T_1)$, β converges from b_0 to b_f , then the link functions $l_2 = \frac{\lambda\beta}{\sqrt{b_0^2 - \beta^2}}$ and $l_1 = \frac{-\lambda\beta}{\sqrt{b_0^2 - \beta^2}}$ converge from $\pm\infty$ to bounded constants/functions; and for $t \in [T_1, \infty)$, $\beta = b_f$, then $\mathcal{F}_2 > 0$ and $\mathcal{F}_1 < 0$ are bounded, which implies that “ $F_{[0, T_1^l]} L_{(T_1^l, T_1)} C_{[T_1, \infty)}$ ” can be ensured.
- 6) For case ④, if $\underline{\delta} = \bar{\delta} = 1$, it is checked from (15) that, for $t \in [0, T_1]$, $\beta = b_f$, then $\mathcal{F}_2 > 0$ and $\mathcal{F}_1 < 0$ are bounded; for $t \in (T_1, T_1^r)$, β increases from b_f to b_0 , then the upper link function l_2 increases from a bounded function to ∞ and the lower link function l_1 decreases from a bounded function to $-\infty$; and for $t \in [T_1^r, \infty)$, $\beta = b_0$, then $\mathcal{F}_2 = \infty$ and $\mathcal{F}_1 = -\infty$, then the “ $C_{[0, T_1]} L_{(T_1, T_1^r)} F_{(T_1^r, \infty)}$ ” form is guaranteed.
- 7) Similarly, the correct choice of $\underline{\delta}, \bar{\delta}, b_0, b_f$ and the decision function κ can lead to similar conclusions for the Cases ⑤ and ⑥ and guarantee the alternating constraint forms.

The proof is completed. $\square$

Remark 4: With the aid of Lemma 1, it is obvious that, for $-\underline{\delta}(0) < \zeta(0) < \bar{\delta}(0)$, if the boundedness of s is guaranteed $\forall t \in [0, \infty)$, by selecting different decision function κ and parameters $\bar{\delta}$ and $\underline{\delta}$, the Type A–Type F constraint forms can be ensured uniformly and the control redesign is not required, which dramatically decreases the difficulty of control design and increases the applicability of the proposed control. In addition, some constraint forms with special $\kappa, \bar{\delta}$, and $\underline{\delta}$ are summarized in Table I.

Remark 5: Here, we also give the following numerical examples to show that different constraint forms can be ensured via choosing $b_0, b_f, \underline{\delta}, \bar{\delta}$, and κ . For example:

- 1) when $\underline{\delta} = \bar{\delta} = 1$ and $\kappa \equiv 1$, one has $\mathcal{F}_2 = -\mathcal{F}_1 = +\infty$, which corresponds to the unconstrained case;
- 2) when $b_0 = 2$, $b_f = 1$, $\underline{\delta} = \sqrt{2}$, $\bar{\delta} = 1$, and $\kappa \equiv 0$, one has $\mathcal{F}_1 = -\lambda$ and $\mathcal{F}_2 = \frac{\lambda}{\sqrt{3}}$. If $\lambda = \sqrt{3}(1 + 0.5 \sin(t))$, the variable $\mathcal{X}$ obeys a persistent and time-varying constraint, i.e., $-\sqrt{3}(1 + 0.5 \sin(t)) < \mathcal{X}(t) < (1 + 0.5 \sin(t))$, $\forall t \in [0, \infty)$;
- 3) when $\underline{\delta} = \bar{\delta} = 1$ and $\kappa(t)$ is chosen as Case ③, one has $\mathcal{F}_2 =$

$$-\mathcal{F}_1 = \begin{cases} +\infty, & t \in [0, T_1^l] \\ +\infty \rightarrow \frac{\lambda b_f}{\sqrt{b_0^2 - b_f^2}}, & t \in (T_1^l, T_1) \\ \frac{\lambda b_f}{\sqrt{b_0^2 - b_f^2}}, & t \in [T_1, \infty). \end{cases}$$
which corresponds to the Type C constraint (free-constrained form).

Therefore, with the above discussion, the normally considered persistent constraint [8], [10], [12], [14], [15] and the deferred constraint in [18] are special cases of the developed method.

D. System Dynamics With s

It is shown from Lemma 1 that the problem of output constraint on $\mathcal{X}$ for the considered system (1) can be converted into the stabilization of the variable s . To this end, here we try to get the system dynamics w.r.t. s . According to (12), the derivative of s is

$$\dot{s} = h_1 \dot{\mathcal{X}} + h_2 \quad (16)$$

with

$$\begin{aligned} h_1 &= \frac{\partial s}{\partial \zeta} \frac{\partial \zeta}{\partial \eta} \frac{\partial \eta}{\partial \mathcal{X}} \\ h_2 &= \frac{\partial s}{\partial \zeta} \left(\frac{\partial \zeta}{\partial \eta} \frac{\partial \eta}{\partial \lambda} \dot{\lambda} + \frac{\partial \zeta}{\partial \beta} \dot{\beta} \right) + \frac{\partial s}{\partial \bar{\delta}} \dot{\bar{\delta}} + \frac{\partial s}{\partial \bar{\delta}} \dot{\bar{\delta}} \\ \frac{\partial s}{\partial \zeta} &= \frac{\bar{\delta} \bar{\delta} + \zeta^2}{(\bar{\delta} + \zeta)^2 (\bar{\delta} - \zeta)^2}, \quad \frac{\partial s}{\partial \eta} = \frac{1}{\beta}, \quad \frac{\partial s}{\partial \beta} = -\frac{\eta}{\beta^2} \\ \frac{\partial \eta}{\partial \mathcal{X}} &= \frac{b_0 \lambda^2}{(\mathcal{X}^2 + \lambda^2) \sqrt{\mathcal{X}^2 + \lambda^2}}, \quad \frac{\partial \eta}{\partial \lambda} = \frac{b_0 \mathcal{X} \lambda}{(\mathcal{X}^2 + \lambda^2) \sqrt{\mathcal{X}^2 + \lambda^2}}. \end{aligned}$$

It can be easily verified that $h_1 > 0$ and h_2 are well defined over the open set $\Omega_\zeta = \{\zeta \in \mathbb{R} \mid -\bar{\delta} < \zeta < \bar{\delta}\}$. In the following, we focus on the design of an adaptive controller to stabilize the transformed nonlinear dynamics as shown in (16).

III. ADAPTIVE CONTROL OF FIRST-ORDER SYSTEMS UNDER IRREGULAR OUTPUT CONSTRAINT

A. Adaptive Control of Time-Invariant Parameter Systems

To gain insight into the main idea of guaranteeing irregular constraints, we consider the following first-order uncertain system:

$$\dot{x} = \theta^\top \phi(x) + u, \quad y = x \quad (17)$$

subject to

$$\Omega_y := \{y \in \mathbb{R} \mid \mathcal{F}_1(t) < y(t) < \mathcal{F}_2(t)\} \quad (18)$$

where $\mathcal{F}_1(t)$ and $\mathcal{F}_2(t)$ are time-varying boundaries possibly formed/imposed from Type A–Type F constraints, $x \in \mathbb{R}$ is the system state, $\theta \in \mathbb{R}^r$ is an unknown system parameter, $\phi(x) \in \mathbb{R}^r$ denotes a smooth nonlinear function, and $u \in \mathbb{R}$ is the control input. Furthermore, the conditions of existence and uniqueness of solution under constraints are assumed to be satisfied.

Let $e = y - y_d$ be the tracking error with y_d being the reference signal, which satisfies the following assumption.

Assumption 1: The reference signal y_d and its first derivative are known, bounded, and piecewise continuous. Furthermore, y_d is within a subset of Ω_y , namely, there exist bounded functions $\mathcal{F}_{2y}$ and $\mathcal{F}_{1y}$ so that $\mathcal{F}_1(t) < \mathcal{F}_{1y}(t) < y_d(t) < \mathcal{F}_{2y}(t) < \mathcal{F}_2(t)$.

Remark 6: Assumption 1 indicates that the reference signal must be within a smaller range than the output constraint. This is because if the evolution of the reference signal is outside the range of the corresponding constraint, it is impossible for the system output to achieve asymptotic tracking. In fact, this is a basic condition for the reference signal in the existing output constraint works [8], [9], [10], [11], [12], [13], [14], [15].

The control objective is to design an adaptive control law for (17) so that not only the boundedness of all signals in the closed-loop systems is bounded and tracking error converges to zero asymptotically, but also the output constraints with Type A–Type F can be handled under a fixed control structure.

1) Transformation: Before designing the adaptive controller, two key steps are required to be carried out. First, we make nonlinear transformations on the system output y and the reference signal y_d

$$s_x = \frac{\zeta_x}{(\bar{\delta} + \zeta_x)(\bar{\delta} - \zeta_x)}, \quad s_d = \frac{\zeta_d}{(\bar{\delta} + \zeta_d)(\bar{\delta} - \zeta_d)}$$

where ζ_x and ζ_d play the identical role with ζ , and $\mathcal{X}$ should be replaced by x and y_d , respectively. For example, $\zeta_d = \frac{\eta_d}{\beta}$ with $\eta_d = \frac{b_0 y_d}{\sqrt{y_d^2 + \lambda^2}}$.

Second, we impose a new coordinate transformation

$$z = s_x - s_d. \quad (19)$$

As $\dot{y}_d$, $\dot{\lambda}$, and $\dot{\kappa}$ are known, it is shown that $\dot{s}_d$ is available and bounded for control design. According to (16), the derivative of z is

$$\dot{z} = h_1 \dot{x} + h_2 - \dot{s}_d = h_1(\theta^\top \phi + u) + h_2 - \dot{s}_d. \quad (20)$$

Remark 7: Here, we state the algebraic relationship between z and the tracking error e . According to the expressions of s_x and s_d , performing lengthy calculations, it is shown that

$$z = \frac{Q}{\beta} \cdot (\zeta_x - \zeta_d) = \frac{b_0 Q}{\beta} \left(\frac{x}{\sqrt{x^2 + \lambda}} - \frac{y_d}{\sqrt{y_d^2 + \lambda}} \right) \quad (21)$$

with $Q = \frac{\bar{\delta} \bar{\delta} + \zeta_x \zeta_d}{P}$ and $P = (\bar{\delta} + \zeta_x)(\bar{\delta} - \zeta_x)(\bar{\delta} + \zeta_d)(\bar{\delta} - \zeta_d)$. Since $P > 0$ and $Q > 0$ for $-\bar{\delta} < \zeta_x < \bar{\delta}$ and $-\bar{\delta} < \zeta_d < \bar{\delta}$, then it is not difficult to get the conclusion that $x \rightarrow y_d$ as $z \rightarrow 0$.

2) Control Design and Analysis: The control law u with the adaptive law for $\hat{\theta}$ is designed as

$$u = -\frac{1}{h_1}(cz + h_2 - \dot{s}_d) - \hat{\theta}^\top \phi(x), \quad \text{with } \dot{\hat{\theta}} = \Gamma z h_1 \phi \quad (22)$$

where $c > 0$ is a design parameter, $\Gamma = \Gamma^\top \in \mathbb{R}^{r \times r}$ is a positive definite matrix, and $\hat{\theta}$ is the parameter estimate of θ . The stability analysis of the closed-loop system under the proposed adaptive controller (22) is summarized in the following theorem.

Theorem 1: Consider the uncertain nonlinear system (17) in the presence of output constraint (18). For any initial condition satisfying $y(0) \in \Omega_y$, under Assumption 1, if the developed control (22) is applied, it is ensured that: 1) all signals in the closed-loop systems are bounded and the tracking error e converges to zero asymptotically, and 2) the various kinds of constraining forms (i.e., Type A–Type F) can be maintained without changing the control structure.

Proof: Substituting the control law (22) into (20), one has

$$\dot{z} = -cz + h_1 \tilde{\theta}^\top \phi \quad (23)$$

with $\tilde{\theta} = \theta - \hat{\theta}$ being the parameter estimate error. We define the Lyapunov function candidate as $V = \frac{1}{2}z^2 + \tilde{\theta}^\top \Gamma^{-1} \tilde{\theta}$, with the aid of adaptive law, the derivative of V is computed as $\dot{V} = -cz^2 \leq 0$.

Now, we first prove that all signals in the closed-loop systems are bounded. Noting that V is bounded, then one has $z \in L_2 \cap L_\infty$ and $\tilde{\theta} \in L_\infty$. According to Assumption 1, one has s_d is bounded, then it follows that s_x is bounded. By utilizing the discussion in Lemma 1, it is ensured that $\Phi(-\bar{\zeta}\beta) \leq y = x \leq \Phi(\bar{\zeta}\beta)$, then the boundedness of system state x is ensured. Apparently, the functions $\phi(x)$, h_1 , and h_2 are all bounded and there exist some positive constants $\underline{h}_1$ and $\bar{h}_1$ so that $0 < \underline{h}_1 \leq h_1 \leq \bar{h}_1 < \infty$, then it follows from (22) and $\dot{z}$ that the control law u , the adaptive law $\dot{\hat{\theta}}$, and $\dot{z}$ are bounded.

Next, we show the asymptotic tracking and output constraint. Noting that $z \in L_2 \cap L_\infty$ and $\dot{z} \in L_\infty$, upon using the Barbalat lemma, it follows that $\lim_{t \rightarrow \infty} z(t) = 0$. With the analysis in Remark 7, one has $\lim_{t \rightarrow \infty} e(t) = 0$ as $\lim_{t \rightarrow \infty} z(t) = 0$. In addition, since s_x is bounded provided $x(0) \in \Omega_y$, then utilizing the discussion in Lemma 1, the output constraint (18) is ensured. The proof is completed. $\square$

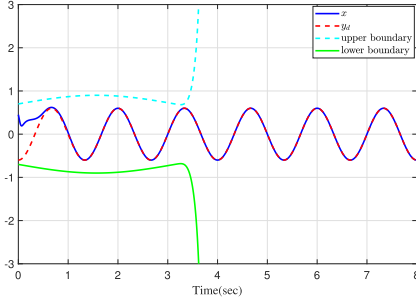


Fig. 2. Evolutions of $x(t)$ and $y_d(t)$ with Type D constraint form under the proposed control.

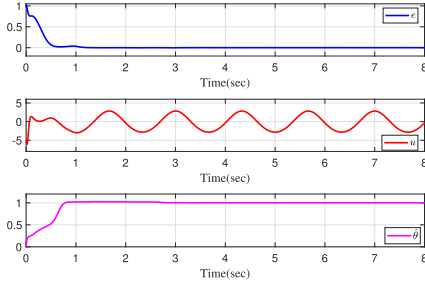


Fig. 3. Evolutions of tracking error e , control input u , and parameter estimate $\hat{\theta}$ under the proposed control.

3) Simulation: To test the effectiveness of the proposed constrained control, we consider the uncertain system (17) with Type D form (i.e., constrained-free form). In this case, the system output is required to be limited in the following set:

$$\begin{cases} \mathcal{F}_1^0(t) < y < \mathcal{F}_2^0(t), & \text{for } t \in [0, 3] \\ -\infty < y < \infty, & \text{for } t \in (3, \infty) \end{cases} \quad (24)$$

with $\mathcal{F}_1^0(t) = -(0.7 + 0.2 \sin(t))$ and $\mathcal{F}_2^0(t) = 0.7 + 0.2 \sin(t)$. By using the link function l_k , we convert the hard constraint (24) into the following smooth soft form, i.e.,

$$\begin{cases} \mathcal{F}_1^0(t) < y < \mathcal{F}_2^0(t), & \text{for } t \in [0, 3] \\ l_1(t) < y < l_2(t), & \text{for } 3 < t < 3.5 \\ -\infty < y < \infty, & \text{for } t \geq 3.5. \end{cases} \quad (25)$$

By choosing the decision function κ as Case ④, link functions l_k as $l_1 = \frac{-\lambda}{\sqrt{b_0^2 - \lambda^2}}$ and $l_2 = \frac{\lambda}{\sqrt{b_0^2 - \lambda^2}}$, and parameters $T_1, T_1^r, b_0, b_f, \underline{\delta}, \bar{\delta}$, and λ as $T_1 = 3, T_1^r = 3.5, b_0 = 1, b_f = 0.25, \underline{\delta} = \bar{\delta} = 1$, and $\lambda = \sqrt{3}(0.7 + 0.2 \sin(t))$, then (25) can be expressed as

$$-\Phi(\underline{\delta}\beta) < y < \Phi(\bar{\delta}\beta), \quad \text{for } t \geq 0.$$

In the simulation, the reference signal is given as $y_d(t) = -0.6 \cos(1.5\pi t)$. The initial conditions are given as $x(0) = 0.45$ and $\hat{\theta}(0) = 0$. The control parameters are selected as $c = 4$ and $\Gamma = 2$. Under the proposed control (22), the simulation results are shown in Figs. 2 and 3. It is observed that, although the reference signal fluctuates rapidly and sometimes approaches the constraining boundaries (e.g., 0, 1.34, 2, and 2.7 s), the system output can asymptotically follow the reference signal and does not violate the irregular “constrained-free” constraint form, which confirms the theoretical discussion.

B. Adaptive Control of Parameter-Varying Systems

Here, we consider a class of first-order systems with time-varying parameters

$$\dot{x} = \theta^\top(t)\phi(x) + u, \quad y = x \quad (26)$$

subject to the output constraint (18), where $\phi(x) \in \mathbb{R}^r$ denotes a smooth nonlinear function and satisfies $\phi(0) = 0$, $\theta(t) \in \mathbb{R}^r$ meets the following assumption.

Assumption 2: The parameter $\theta(t)$ is piecewise continuous and $\theta(t) \in \Theta_0 \forall t \geq 0$, where Θ_0 is a compact set.

Since the parameter $\theta(t)$ is unknown and time-varying, it can change rapidly, which means that the derivative of θ can become infinite, so that the traditional adaptive controls [3], [7], which normally work with constant parameters, are no longer applicable. Here, borrowed from the congelation of variable method [21], we develop an adaptive control method to maintain the output constraint for parameter-varying systems.

In this section, we consider the case $y_d = 0$, then $s_d = 0$ and $z = s_x$. Therefore, (20) can be rewritten as

$$\dot{z} = h_1(\theta^\top \phi + u) + h_2. \quad (27)$$

To handle the (fast) time-varying parameter θ , we rewrite θ as

$$\theta = \hat{\theta} + \Delta_\theta + \ell_\theta - \hat{\theta}$$

where $\hat{\theta}$ is the parameter estimate, and $\Delta_\theta = \theta - \ell_\theta$ with ℓ_θ being a constant vector that can be regarded as the average of $\theta(t)$. Since $\theta(t)$ is time-varying and bounded, it is easily shown that there exists a constant δ_θ so that $\|\Delta_\theta\| \leq \delta_\theta$.

Therefore, (27) further becomes

$$\dot{z} = h_1 \left[\hat{\theta} + \Delta_\theta + (\ell_\theta - \hat{\theta}) \right]^\top \phi(x) + h_1 u + h_2. \quad (28)$$

Before developing the adaptive controller, we give some special functions. As $\phi(0) = 0$, then $\phi(x)$ can be written as $\phi(x) = x\varphi(x)$. In addition, it is shown from (12) that

$$x = \Psi(\cdot)s_x$$

with $\Psi = \frac{1}{b_0} \sqrt{x^2 + \lambda^2} \beta(\underline{\delta} + \zeta_x)(\bar{\delta} - \zeta_x)$.

1) Control Design and Stability Analysis: The adaptive controller is designed as

$$\begin{cases} u = -\frac{1}{h_1}(cz + h_2) - \hat{\theta}^\top \phi + u_r \\ u_r = -\frac{1}{2}\hat{\delta}_\theta(\epsilon_\theta z + \frac{1}{\epsilon_\theta}\Psi^2\varphi^2 z) \\ \dot{\hat{\theta}} = \Gamma\phi h_1 z \\ \dot{\hat{\delta}}_\theta = \frac{\gamma_\theta}{2}(\epsilon_\theta h_1 z^2 + \frac{1}{\epsilon_\theta}h_1\phi^2) \end{cases} \quad (29)$$

where $c > 0$, $\epsilon_\theta > 0$, and $\gamma_\theta > 0$ are design parameters, $\Gamma = \Gamma^\top \in \mathbb{R}^{r \times r}$ is a positive definite matrix, $\hat{\theta}$ and $\hat{\delta}_\theta$ are the parameter estimates of θ and δ_θ , respectively.

Theorem 2: Consider the uncertain nonlinear system (26) with time-varying parameter. For any initial condition satisfying $y(0) \in \Omega_y$, under Assumptions 1 and 2 if the developed control (29) is proposed, it ensures that: 1) all signals in the closed-loop systems are bounded and the system state converges to zero asymptotically, and 2) the various kinds of constraint forms (i.e., Type A–Type F) can be maintained.

Proof: Substituting the controller u defined in (29) into (28), one has

$$\dot{z} = h_1 \Delta_\theta^\top \varphi \Psi z + h_1(\ell_\theta - \hat{\theta})^\top \phi(x) - cz + h_1 u_r. \quad (30)$$

Then, the derivative of quadratic function $V_1 = \frac{1}{2}z^2 + \frac{1}{2}(\ell_\theta - \hat{\theta})^\top \Gamma^{-1}(\ell_\theta - \hat{\theta})$ along (30) yields

$$\dot{V}_1 = h_1 \Delta_\theta^\top \varphi \Psi z^2 - cz^2 + h_1 z u_r \quad (31)$$

where $\hat{\theta} = \Gamma\phi h_1 z$ is used. Noting that $h_1 \Delta_\theta^\top \varphi \Psi z^2$ can be upper bounded by

$$\begin{aligned} h_1 \Delta_\theta^\top \varphi \Psi z^2 &\leq \sqrt{h_1}|z|\sqrt{h_1}\delta_\theta|\varphi||\Psi||z| \\ &\leq \frac{1}{2}\epsilon_\theta h_1 \delta_\theta z^2 + \frac{1}{2\epsilon_\theta} h_1 \delta_\theta \Psi^2 \varphi^2 z^2. \end{aligned}$$

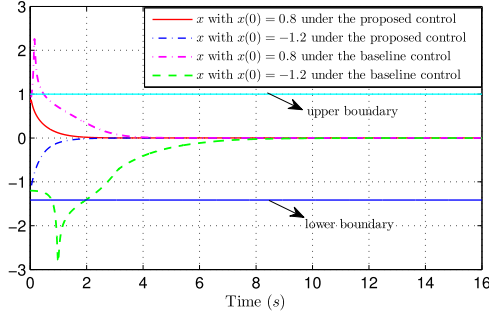


Fig. 4. Evolution of $y(t)$ with persistent output constraints under the proposed control and the baseline control in [21].

Thus, with the aid of u_r , (31) becomes

$$\dot{V}_1 \leq -cz^2 + \frac{\tilde{\delta}_\theta}{2} \left(\epsilon_\theta h_1 z^2 + \frac{1}{\epsilon_\theta} h_1 \phi^2 \right) \quad (32)$$

with $\tilde{\delta}_\theta = \delta_\theta - \hat{\delta}_\theta$ is the parameter estimate error.

Choosing the Lyapunov function candidate as $V = V_1 + \frac{1}{2\gamma_\theta} \tilde{\delta}_\theta^2$, then its derivative, together with the adaptive law $\dot{\hat{\delta}}_\theta$, is $\dot{V} = -cz^2 \leq 0$. The proofs on the boundedness of closed-loop signals, the asymptotic convergence of x , and the maintenance of output constraints are similar to that in Theorem 1, then they are omitted.

To test the effectiveness of the proposed constrained control, we choose the unconstrained adaptive control in [21] as the baseline controller.² In the simulation, we employ an uncertain system (26) with $\theta(t) = 2 + \text{sign}(\sin(t))$ and $\phi(x) = x^2$. Moreover, we consider the persistent constraint, i.e., $-\sqrt{2} < x(t) < 1 \forall t \geq 0$. Under such a constraint form, the decision function is chosen as $\kappa = 0$, the other parameters are given as $b_0 = 2$, $b_f = 1$, $\lambda = 1$, $\bar{\delta} = \sqrt{2}$, and $\underline{\delta} = \frac{2\sqrt{2}}{\sqrt{3}}$. The design parameters in both control schemes are given as: $c = 0.7$, $\epsilon_\delta = 1$, $\Gamma = 0.2$, and $\gamma_\delta = 0.2$. The initial conditions are given as: $x(0) = 0.8/-1.2$, $\hat{\theta}(0) = 0$, and $\hat{\delta}_\theta(0) = 0$. The simulation result is shown in Fig. 4, from which it is seen that, although the parameter $\theta(t)$ is fast time-varying and nondifferentiable, the proposed control not only ensures the asymptotic result but also the constraint is not transgressed.

IV. ADAPTIVE CONTROL OF STRICT-FEEDBACK SYSTEMS UNDER IRREGULAR OUTPUT CONSTRAINT

1) System Model: Motivated by the result in Section III, we extend the proposed adaptive control algorithm to the following uncertain strict-feedback nonlinear systems [3]:

$$\begin{cases} \dot{x}_j = \theta^\top \phi_k(\bar{x}_k) + x_{j+1}, & j = 1, 2, \dots, n-1 \\ \dot{x}_n = \theta^\top \phi_n(\bar{x}_n) + u \\ y = x_1 \end{cases} \quad (33)$$

subject to $\Omega_y := \{y \in \mathbb{R} | \mathcal{F}_1(t) < y(t) < \mathcal{F}_2(t)\}$. In (33), $x_i \in \mathbb{R}$, $i = 1, \dots, n$, is the system state with $\bar{x}_i = [x_1, \dots, x_i]^\top \in \mathbb{R}^i$, $y \in \mathbb{R}$ is the system output, $u \in \mathbb{R}$ is the control signal, $\theta \in \mathbb{R}^r$ is an unknown constant parameter vector, and $\phi_i(\bar{x}_i) \in \mathbb{R}^r$ is a smooth nonlinear function, which is available for control design.

The objective of this section is to design an adaptive control law $u(t)$ for (33) so that 1) the boundedness of all the signals and tracking error $e(t) = y(t) - y_d(t)$ asymptotically approaching zero are ensured, and

²The adaptive law in [21] is $u = -cx - \hat{\theta}^\top \phi - \frac{1}{2} \epsilon_\theta \hat{\delta}_\theta x - \frac{1}{2\epsilon_\theta} \hat{\delta}_\theta \phi^2 x$ with $\dot{\hat{\theta}} = \Gamma x \phi$ and $\dot{\hat{\delta}}_\theta = \frac{\gamma_\theta}{2} (\epsilon_\theta x^2 + \frac{1}{\epsilon_\theta} \phi^2)$.

TABLE II
ADAPTIVE BACKSTEPPING CONTROLLER

Virtual Control Laws: ($i = 3, \dots, n$)	
$\alpha_1 = -\frac{1}{h_1} (c_1 z_1 + h_2 - \dot{s}_d) - \hat{\theta}^\top \phi_1, \quad c_1 > 0$ $\alpha_2 = -h_1 z_1 - c_2 z_2 - \hat{\theta}^\top \omega_2 + \frac{\partial \alpha_1}{\partial \theta} \Gamma \tau_2 - \sigma_2, \quad c_2 > 0$ $\alpha_i = -z_{i-1} - c_i z_i - \sigma_i + \frac{\partial \alpha_{i-1}}{\partial \theta} \Gamma \tau_i$ $\quad + \left(\sum_{k=1}^{i-2} z_{k+1} \frac{\partial \alpha_k}{\partial \theta} \Gamma - \hat{\theta}^\top \right) \omega_i, \quad c_i > 0$ $\tau_1 = z_1 h_1 \phi_1, \tau_j = \tau_{j-1} + z_j \omega_j, \quad j = 2, \dots, n$ $\omega_1 = h_1 \phi_1, \omega_j = \phi_j - \sum_{k=1}^{j-1} \frac{\partial \alpha_{j-1}}{\partial x_k} \phi_k$ $\sigma_j = -\sum_{k=1}^{j-1} \frac{\partial \alpha_{j-1}}{\partial \beta^{(k)}} x_{k+1} - \sum_{k=0}^{j-1} \frac{\partial \alpha_{j-1}}{\partial y_d^{(k)}} y_d^{(k+1)}$ $\quad - \sum_{k=0}^{j-1} \frac{\partial \alpha_{j-1}}{\partial \beta^{(k)}} \beta^{(k+1)} - \sum_{k=0}^{j-1} \frac{\partial \alpha_{j-1}}{\partial \lambda^{(k)}} \lambda^{(k+1)}$ $\quad - \sum_{k=0}^{j-1} \frac{\partial \alpha_{j-1}}{\partial \delta^{(k)}} \delta^{(k+1)} - \sum_{k=0}^{j-1} \frac{\partial \alpha_{j-1}}{\partial \bar{\delta}^{(k)}} \bar{\delta}^{(k+1)}.$	
Actual Control Law: $u = \alpha_n$	(T2.2)
Parameter Update Law: $\dot{\hat{\theta}} = \Gamma \tau_n$	(T2.3)

2) the imposed irregular output constraints can be handled uniformly without changing the control structure.

Assumption 3: The reference signal y_d and $y_d^{(k)}$, $k = 1, \dots, n$, are known, bounded, and piecewise continuous. Furthermore, y_d is within a subset of Ω_y , namely, there exist bounded functions $\mathcal{F}_{2y}$ and $\mathcal{F}_{1y}$ so that $\mathcal{F}_1(t) < \mathcal{F}_{1y}(t) < y_d(t) < \mathcal{F}_{2y}(t) < \mathcal{F}_2(t)$.

2) Control Design and Analysis: By recursively following the backstepping design procedure [3], we define the following coordinate transformation:

$$z_1 = s_x - s_d, \quad z_i = x_i - \alpha_{i-1}, \quad i = 2, \dots, n \quad (34)$$

where α_{i-1} is the virtual control law. Under such a transformation, the adaptive constrained controller for (33) can be obtained, as summarized in Table II. In addition, we can obtain the results in the following theorem.

Theorem 3: Consider the uncertain nonlinear system (33) satisfying Assumption 3. Under the virtual controllers (T2.1), the actual controller (T2.2), and the parameter estimate (T2.3), the following is ensured:

- 1) all signals in the closed-loop systems are bounded;
- 2) the tracking error asymptotically converges to zero;
- 3) the irregular output constraint is maintained.

Proof: From the virtual/actual control design, it is shown that the derivative of the Lyapunov function candidate as $V = \frac{1}{2} \sum_{k=1}^n z_k^2 + \frac{1}{2} \tilde{\theta}^\top \Gamma^{-1} \tilde{\theta}$, with $\tilde{\theta} = \theta - \hat{\theta}$, is $\dot{V} = -\sum_{k=1}^n z_k^2 \leq 0$. Similar to the discussion in the above sections and the analysis in [3], the boundedness of all signals with asymptotic tracking can be ensured and the violation of output constraint can be prevented.

Remark 8: The purpose of introducing T_i^l and T_i^r is to construct the link function and convert the hard constraint into a soft one, but the price of the link function is that it is inevitable to impose an additional constraint on the system output during the interval (T_i^l, T_i) or (T_i, T_i^r) . If the distance between T_1^l and T_1 (or T_1^r and T_1) is arbitrarily small, the soft constraint form can reproduce the actual hard constraint form sufficiently accurately, and the negative effects on the system output and the control algorithm can be almost ignored.

3) Simulation: To test the advantages of our proposed method, we consider the following robotic manipulator system:

$$\ddot{x} + B\dot{x} + MGL \sin(x) = u$$

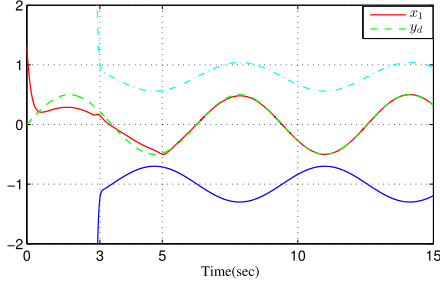


Fig. 5. Evolutions of x_1 and y_d with Type C constraint.

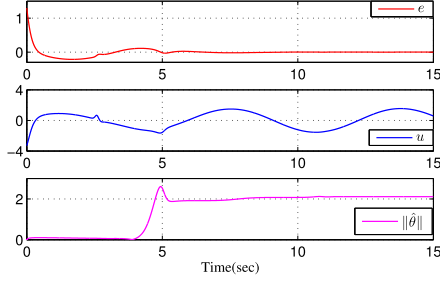


Fig. 6. Evolutions of tracking error e , control input u , and parameter estimate $\hat{\theta}$ under the proposed control.

where $x \in \mathbb{R}$ and $\dot{x} \in \mathbb{R}$ are the angle position and angular velocity of the rigid link, respectively. The definitions of B and MGL can be found in [20].

Let $x_1 = x$ and $x_2 = \dot{x}$. The system output $y = x_1$ is required to satisfy the soft Type C constraint (“ $F_{[0, T_1^l]} L_{(T_1^l, T_1)} C_{[T_1, \infty)}$ ”)

$$\begin{cases} -\infty < x_1(t) < \infty, & t \in [0, T_1^l] \\ -\frac{\lambda \delta \beta}{\sqrt{b_0^2 - (\delta \beta)^2}} < x_1(t) < \frac{\lambda \bar{\delta} \beta}{\sqrt{b_0^2 - (\delta \beta)^2}}, & t \in (T_1^l, T_1) \\ \mathcal{F}_1^1(t) < x_1(t) < \mathcal{F}_2^1(t), & t \in [T_1, \infty) \end{cases}$$

where $\mathcal{F}_1^1(t) = -1 - 0.3 \sin(t)$, $\mathcal{F}_2^1(t) = 0.8 + 0.24 \sin(t)$, $b_0 = 2$, $b_f = 1$, $T_1^l = 2.7$, $T_1 = 3$, and $\lambda = 2 + 0.6 \sin(t)$, $\bar{\delta} = (1 - \frac{4}{\sqrt{29}})\kappa + \frac{4}{\sqrt{29}}$, and $\delta = (1 - \frac{2}{\sqrt{5}})\kappa + \frac{2}{\sqrt{5}}$ with κ in Case (3).

In the simulation, the reference signal is chosen as $y_d(t) = 0.5 \sin(t)$ (rad). The initial values of system states and parameter estimates are set as $x_1(0) = 1.3$ (rad), $x_2(0) = -1$ (rad/s), and $\hat{\theta}(0) = [0, 0]^T$. The design parameters are chosen as $c_1 = 1$, $c_2 = 4$, and $\Gamma = \text{diag}\{1.5\}$. The simulation results are presented in Figs. 5 and 6. The trajectories of system output x_1 and the reference signal y_d under Type C constraint are plotted in Fig. 5, from which it is shown that, under the proposed control, the cases with “free-constrained” form on the system output can be handled, which confirms the theoretical prediction. Furthermore, the evolutions of tracking error e , control input u , and the norm of parameter estimate $\|\hat{\theta}\|$ are plotted in Fig. 6, which are all bounded for $t \in [0, \infty)$.

V. CONCLUSION

A unified adaptive control strategy has been developed for uncertain nonlinear systems subject to irregular output constraints. First, by proposing a link function, we convert the hard form into a class of soft constraints. Second, we construct a series of nonlinear function-based transformations (especially the soft function and decision function), so that the guarantee of different constraint types is transformed into the selection of the decision function and the main design parameters,

eliminating the need to redesign the controller and reanalyze the stability, and providing a more powerful and general adaptive design procedure for dealing with output constraints. However, it is worth noting that in this note only the output constraint is considered, whereas full-state constraints normally exist in many practical applications. Therefore, extending the proposed method to irregular full-state constraints for strict-feedback uncertain systems represents an interesting topic for future research.

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